

GNN-Augmented BP Architectures for QLDPC Decoding

Jothiradithya Sai Konduru¹, Anish Chedalla¹, and Nithin Raveendran²

¹Paradise Valley High School, Phoenix, AZ, USA

²Department of Electrical and Computer Engineering, University of Arizona, Tucson, AZ, USA

Bivariate bicycle (BB) quantum low-density parity-check (QLDPC) codes combine finite encoding rate with low-weight checks, but their loopy Tanner graphs remain challenging for iterative decoding. Learned message-passing can improve belief propagation (BP), although the measured gain depends strongly on the BP configuration used for training and evaluation. We study this issue using a Tanner-graph neural network (TannerGNN) that produces qubit-wise log-likelihood corrections for BP under a biased CSS component-noise model. On the $[[72, 12, 6]]$ BB code, a model trained with 10-iteration flooding BP reduces the logical error rate (LER) ~~from 0.0990 to 0.0838 on 8 of that decoder by~~ 12.5% $\pm$ 0.6% on 20,000 held-out syndromes at $p = 0.04$ and $\eta = 20$, ~~a 15.4% relative reduction (paired McNemar $p < 10^{-15}$ (five training seeds; paired McNemar $p < 10^{-36}$ for every seed).~~ This gain does not transfer to a stronger serial min-sum BP-OSD decoder: the learned correction ~~is statistically indistinguishable at lower tested error rates and degrades performance at $p \geq 0.04$ never improves it and is significantly worse at $p \geq 0.03$. A paired failure-set analysis shows that the failure sets of the two decoders are nested and that most shots the GNN repairs are already decoded by BP-OSD.~~ Independent baseline experiments further show that schedule, BP update rule, iteration count, and OSD configuration can change LER by orders of magnitude. These settings must therefore be specified and controlled when learned decoders are compared. We also establish that unclipped min-sum BP is invariant, in its hard decisions, to a uniform positive rescaling of all channel LLRs

Nithin Raveendran: nithin@arizona.edu

within a CSS component. Thus, changing only the magnitude of a uniform prior cannot make the underlying min-sum BP adaptive, although a learned network may still use global context through other pathways. Finally, synthetic experiments with heterogeneous per-qubit error rates show a large gap between mean-prior and oracle-prior decoding, motivating spatially resolved calibration side information rather than a single global noise scalar. The experiments show that an improvement over one BP baseline does not by itself establish an improvement over a stronger decoder, and that spatial calibration information is qualitatively different from a single global noise estimate.

1 Introduction

Bivariate bicycle (BB) codes provide a concrete finite-length setting in which QLDPC codes combine non-vanishing encoding rate, weight-six stabilizer checks, and syndrome-extraction circuits compatible with low-degree connectivity [1]. IBM's current fault-tolerant architecture is explicitly based on BB codes, making the finite-length decoding behavior of this family practically relevant as well as theoretically interesting [2]. The canonical examples include the $[[72, 12, 6]]$, $[[144, 12, 12]]$, and $[[288, 12, 18]]$ codes studied here.

The decoding bottleneck is not the absence of efficient iterative algorithms, but their finite-length failure modes. BP is attractive because its message updates are local and parallelizable, yet short cycles and degeneracy in QLDPC Tanner graphs create correlated messages and non-convergent or incorrect fixed points [3]. OSD post-processing can recover many BP failures, while recent learned decoders use graph neural networks (GNNs) either to replace BP or to inter-

vene between BP stages [4, 5]. These approaches make the choice of baseline unusually consequential: a neural correction trained against one BP schedule or stopping rule need not address the residual failures of a stronger decoder.

A second issue is adaptivity. Hardware noise is neither spatially nor temporally uniform, and calibration information can therefore be useful to a decoder. However, three distinct sources of information should not be conflated: a global estimate of the physical error rate, spatially resolved per-qubit calibration data, and syndrome history. A decoder may respond differently to each. In particular, changing the magnitude of a uniform channel prior has a restricted effect on min-sum BP, whereas per-qubit priors can change the relative reliability ordering that drives both BP and OSD.

We study a GNN-augmented BP pipeline with two questions in view: *what gain does the learned correction provide relative to the BP decoder used in training, and what additional information is needed for noise adaptation?* We compare decoder configurations, training and evaluation baselines, and different forms of calibration information to isolate the source of the observed performance changes.

1.1 Contributions

Our main findings are:

1. **A held-out GNN gain over its training-aligned BP baseline.** On $[[72, 12, 6]]$ at $p = 0.04$ and $\eta = 20$, TANNERGNN+flooding BP reduces LER ~~from 0.0990 to 0.0838 on 8 by 12.5% $\pm$ 0.6% on 20,000 held-out syndromes, corresponding to a 15.4% relative reduction (McNemar $p < 10^{-15}$ over five training seeds (McNemar $p < 10^{-36}$ for every seed)).~~ The ~~model is trained only on $p \in [0.02, 0.03]$, so this test is also outside the training error-rate range~~ training rates span $p \in [0.005, 0.045]$; at $p = 0.06$, outside that range, the reduction is $10.6\% \pm 0.7\%$ (Sec. 5.3).
2. **Baseline configuration and alignment can dominate the apparent learned gain.** Serial min-sum BP with OSD post-processing is substantially stronger than the flooding BP used in GNN training. The same learned correction does not improve this stronger decoder and is ~~worse at the two highest tested error~~

~~rates~~ significantly worse from $p = 0.03$ on. A paired failure-set analysis shows that 93% of BP-OSD failures are also BP failures and that most shots the GNN repairs are already decoded by BP-OSD. Separate experiments on $[[288, 12, 18]]$ show order-of-magnitude sensitivity to BP schedule, update rule, iteration count, and OSD settings. The training decoder and evaluation decoder are therefore part of the experimental condition rather than interchangeable implementations of “BP” (Secs. 5.1 and 5.4).

3. **Global prior scaling and spatial calibration carry different information.** We prove that hard decisions of unclipped min-sum BP are invariant to a uniform positive rescaling of the channel LLRs within a CSS component. This result limits adaptivity obtained only by changing a uniform BP prior; it does not rule out a learned network using a global noise scalar as context. In synthetic heterogeneous-noise experiments, providing the true per-qubit rates to BP-OSD yields large oracle improvements, motivating spatially resolved calibration inputs. Because code size and operating point co-vary in these experiments, we do not attribute the cross-code trend to distance alone (Secs. 5.2 and 5.5).

2 Background

2.1 CSS codes and bivariate bicycle construction

A CSS code [6, 7] is defined by matrices $\mathbf{H}_x, \mathbf{H}_z \in \mathbb{F}_2^{m \times n}$ with $\mathbf{H}_x \mathbf{H}_z^\top = \mathbf{0}$. The syndrome of an error (e_x, e_z) decomposes as $s_x = \mathbf{H}_x e_z$, $s_z = \mathbf{H}_z e_x$ over $\mathbb{F}_2$, so X - and Z -type errors are decoded independently. This CSS independence is structurally important for the results of Sec. 5.2.

Bivariate bicycle codes [1] are constructed over the group algebra $\mathbb{F}_2[x, y]/(x^\ell - 1, y^m - 1)$. Polynomials $A(x, y)$ and $B(x, y)$ yield circulant block matrices:

$$\mathbf{H}_x = [\mathbf{A} \mid \mathbf{B}], \quad \mathbf{H}_z = [\mathbf{B}^\top \mid \mathbf{A}^\top]. \quad (1)$$

We study three codes (Table 1).

2.2 Belief propagation and its failure modes

Min-sum BP on a CSS Tanner graph iterates variable-to-check and check-to-variable messages

Table 1: Bivariate bicycle codes studied in this work, with generator polynomials from [1]; $\mathbf{H}_x$ and $\mathbf{H}_z$ follow Eq. (1) over $\mathbb{F}_2[x, y]/(x^\ell - 1, y^m - 1)$.

Code	Name	n	k	d	Rate	ℓ	m	$A(x, y)$	$B(x, y)$
$[[72, 12, 6]]$	—	72	12	6	1/6	6	6	$x^3 + y + y^2$	$y^3 + x + x^2$
$[[144, 12, 12]]$	gross	144	12	12	1/12	12	6	$x^3 + y + y^2$	$y^3 + x + x^2$
$[[288, 12, 18]]$	two-gross	288	12	18	1/24	12	12	$x^3 + y^2 + y^7$	$y^3 + x + x^2$

for T steps. For a check c_j with measured syndrome bit s_j and a neighboring variable v_i ,

$$\mu_{v_i \rightarrow c_j}^{(t)} = \Lambda_i + \sum_{j' \in \mathcal{N}(i) \setminus j} \mu_{c_{j'} \rightarrow v_i}^{(t-1)}, \quad (2)$$

$$\begin{aligned} \mu_{c_j \rightarrow v_i}^{(t)} &= (-1)^{s_j} \prod_{i' \in \mathcal{N}(j) \setminus i} \text{sgn}(\mu_{v_{i'} \rightarrow c_j}^{(t)}) \\ &\times \min_{i' \in \mathcal{N}(j) \setminus i} |\mu_{v_{i'} \rightarrow c_j}^{(t)}|. \end{aligned} \quad (3)$$

The channel LLR for qubit i under Z -biased noise is $\Lambda_{z,i} = \ln((1 - p_z)/p_z)$, $p_z = p\eta/(\eta + 1)$. CSS decoding separates X and Z components, processing each with its respective parity-check matrix.

The Tanner graphs of the finite-length BB codes considered here contain many short cycles. Message reuse around these cycles creates statistical dependencies that are absent on a tree and can lead BP to oscillate, converge to a syndrome-inconsistent estimate, or converge to the wrong logical class. We therefore treat short-cycle structure as a finite-length decoder mechanism to be measured, rather than attributing every cycle directly to the CSS orthogonality condition. Post-processing methods including ordered-statistics decoding (OSD) [8] recover many such failures but add $O(n^3)$ worst-case complexity. As shown in Sec. 5.1, the specific OSD configuration—schedule, base method, and iteration count—can change LER by more than two orders of magnitude on the same code. We therefore report these settings explicitly in the comparisons below.

2.3 Noise models

We use a Z -biased CSS component-noise model with $\eta = 20$:

$$p_z = \frac{p\eta}{\eta + 1}, \quad p_x = \frac{p}{\eta + 1}. \quad (4)$$

Each data qubit independently suffers a Z flip with probability p_z and an X flip with probability

p_x . No separate Y channel is sampled: a qubit carries a Y error only if it receives both flips, which happens with probability $p_x p_z \approx p^2/22$, so this is not depolarizing noise. Syndromes are $s_x = \mathbf{H}_x e_z$ and $s_z = \mathbf{H}_z e_x$, and the two components are decoded separately. For temporal drift experiments we apply three modulation models to the base rate p : sinusoidal ($p(t) = p + A \sin(2\pi t/T)$), Ornstein-Uhlenbeck, and random telegraph noise (RTN), all motivated by TLS physics [9, 10]. The decoder receives no direct access to instantaneous noise parameters in any experiment.

3 The ADAPTIVEBP Framework

3.1 TANNERGNN

TANNERGNN operates on the unified CSS Tanner graph, with edges distinguished by type $\tau \in \{X\text{-check}, Z\text{-check}\}$. Each variable node v_i is initialised with node receives a 4-dimensional feature vector $\mathbf{x}_i = [\Lambda_i, f_i, 1, 0]$, where f_i is the fraction of neighbouring checks that are violated; each check whose first entry is a value and whose last three entries one-hot encode the node type: a variable node v_i receives $\mathbf{x}_i = [\ell, 1, 0, 0]$, where $\ell = \ln((1 - p)/p)$ is the prior LLR, and an X -check (Z -check) node c_j receives $\mathbf{x}_j = [0, s_j, 0, 1]$ ($[s_j, 0, 1, 0]$) ($[s_j, 0, 0, 1]$), where s_j is its syndrome bit.

An input projection maps features to hidden dimension $d_h = 64$ via $\mathbf{h}_i^{(0)} = \text{LayerNorm}(\mathbf{W}_{\text{proj}} \mathbf{x}_i) d_h$ ($d_h = 32$ for the model of Sec. 5.3) via $\mathbf{h}_i^{(0)} = \text{ReLU}(\text{LayerNorm}(\mathbf{W}_{\text{proj}} \mathbf{x}_i))$. Each of $L = 3$ message-passing layers applies separate edge-type MLPs, optional attention weighting,

and a GRU node update:

$$\mathbf{m}_i^{(\ell)} = \sum_{j \in \mathcal{N}(i)} \text{MLP}_{e,\tau(i,j)}^{(\ell)}([\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}]), \quad (5)$$

$$\mathbf{h}_i^{(\ell+1)} = \text{GRU}(\mathbf{h}_i^{(\ell)}, \mathbf{m}_i^{(\ell)}). \quad (6)$$

A readout MLP maps each variable node's final hidden state to a per-qubit LLR correction: $\Delta_i = \text{MLP}_{\text{read}}(\mathbf{h}_i^{(L)})$.

Remark 1 (Architectural limitation under biased noise). *The current implementation applies the same scalar Δ_i to both $\Lambda_{z,i}$ and $\Lambda_{x,i}$, and computes GNN input features from ~~their average~~ a single prior LLR ℓ . Under $\eta = 20$, $\Lambda_{z,i}$ and $\Lambda_{x,i}$ differ substantially (factor ≈ 1.7 in LLR magnitude at $p = 0.04$), so a single correction cannot simultaneously be optimal for both error types. Separate Z- and X-readout heads provide a direct way to remove this coupling and should be tested under the same biased-noise conditions.*

3.2 FiLM conditioning

Feature-wise Linear Modulation (FiLM) [11] conditions each GNN layer on an external context. A small generator network maps the syndrome-weight estimate $\hat{p} = \|\mathbf{s}\|_1/(m_x + m_z)$ to per-layer affine parameters $(\gamma^{(\ell)}, \beta^{(\ell)}) = \text{FiLMGen}(\hat{p})$, modulating hidden activations as $\tilde{\mathbf{h}}^{(\ell)} = \gamma^{(\ell)} \odot \mathbf{h}^{(\ell)} + \beta^{(\ell)}$. This conditioning pathway is designed to let a single model respond to global changes in the noise level without retraining. It should not be confused with spatial calibration: $\hat{p}$ is one scalar shared by the entire graph and therefore contains no direct per-qubit reliability information. Section 5.2 establishes a narrower result: uniform rescaling of the channel prior cannot change the hard decisions of the underlying unclipped min-sum BP decoder. That invariance does not, by itself, imply that a GNN cannot use a global scalar as contextual information.

3.3 Neural BP and interleaved pipeline

Differentiable BP generalises (2)–(3) with learnable per-iteration scaling weights $w_{\text{ch}}^{(t)}, w_{\text{vtr}}^{(t)}, w_{\text{ctv}}^{(t)} = \text{softplus}(w_{\text{raw}}^{(t)})$ (softplus ensures positivity; $w_{\text{raw}} = 0$ at initialisation gives $w \approx \ln 2 \approx 0.693$, recovering near-standard BP).

Algorithm 1 gives the interleaved pipeline. The GNN intervenes twice: before BP to correct channel priors, and after T_1 iterations using the intermediate posterior information. Qubits with

Algorithm 1: Interleaved GNN-BP Decoding

Input: $\mathbf{s}$, channel LLR Λ , noise estimate $\hat{p}$

Output: Error estimate $\hat{\mathbf{e}}$

$(\gamma, \beta) \leftarrow \text{FiLMGen}(\hat{p})$ // Conditioning

$\Delta^{(1)} \leftarrow \text{TANNERGNN}(\Lambda, \mathbf{s}, \gamma, \beta)$

$\Lambda' \leftarrow \Lambda + \Delta^{(1)}$ // Prior correction

$\Lambda_{\text{mid}} \leftarrow \text{NeuralBP}(\Lambda', \mathbf{H}, T_1)$

$p_i^{\text{mid}} \leftarrow \sigma(-\Lambda_{\text{mid},i}) \forall i$

$\Delta^{(2)} \leftarrow \text{TANNERGNN}(\Lambda_{\text{mid}}, p^{\text{mid}}, \mathbf{s}, \gamma, \beta)$

$\Lambda'' \leftarrow \Lambda_{\text{mid}} + \Delta^{(2)}$ // Mid correction

$\Lambda_{\text{final}} \leftarrow \text{NeuralBP}(\Lambda'', \mathbf{H}, T_2)$

$\hat{e}_i \leftarrow \mathbb{1}[\Lambda_{\text{final},i} < 0] \forall i$

return $\hat{\mathbf{e}}$

$p_i^{\text{mid}} \approx 0.5$ have low posterior confidence and can receive a second learned correction. Both correction stages are trained jointly through the differentiable BP unroll.

3.4 Training

~~Training uses focal loss [12] with $\alpha = 0.25$, $\gamma = 2.0$ for class-imbalanced qubit errors, together with a syndrome-consistency term. The noise range is experiment dependent; in particular, the positive held-out result~~ For a shot with prior LLR $\ell = \ln((1-p)/p)$ on every qubit, the GNN outputs a correction Δ_j for each qubit j , giving the corrected error probability $q_j = \sigma(-(\ell + \Delta_j))$. The model of Sec. 5.3 uses training rates $p \in [0.02, 0.03]$. **Reconcile this with the earlier training configuration $p \sim \mathcal{U}$** ~~Key hyperparameters are summarized in~~ is trained with the focal loss [12] against the true Z error e_z .

$$\mathcal{L} = -\frac{1}{Bn} \sum_{b=1}^B \sum_{j=1}^n \alpha_t (1 - q_t)^\gamma \log q_t, \quad (7)$$

where $q_t = q_j$ and $\alpha_t = \alpha$ if $e_{z,j} = 1$, and $q_t = 1 - q_j$ and $\alpha_t = 1 - \alpha$ otherwise, with $\alpha = 0.25$ and $\gamma = 2$; no syndrome-consistency term is used, and the corrected LLRs $\ell + \Delta_j$ are passed to both CSS components. The training data are 8,000 shots from two sinusoidally drifting streams, $p(t) = p_0 + 0.015 \sin(2\pi t/500)$ with $p_0 \in \{0.02, 0.03\}$, so the training rates span $p \in [0.005, 0.045]$. The FiLM and interleaved variants of Secs. 3.2–3.3 are

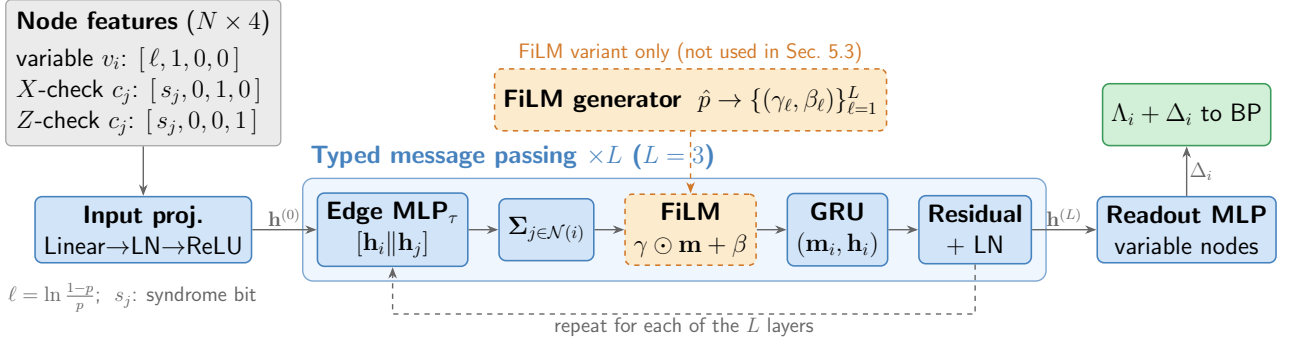


Figure 1: TANNERGNN architecture. Variable and check nodes exchange typed messages over $L = 3$ layers via edge-type-specific MLPs; a GRU update aggregates messages with a residual connection and layer normalisation follows at each node, and a readout MLP on the variable nodes gives the LLR corrections. In the FiLM modulation variant (not shown dashed), FiLM modulation scales hidden activations with syndrome-derived conditioning the aggregated messages at each layer; the model of Sec. 5.3 does not use FiLM.

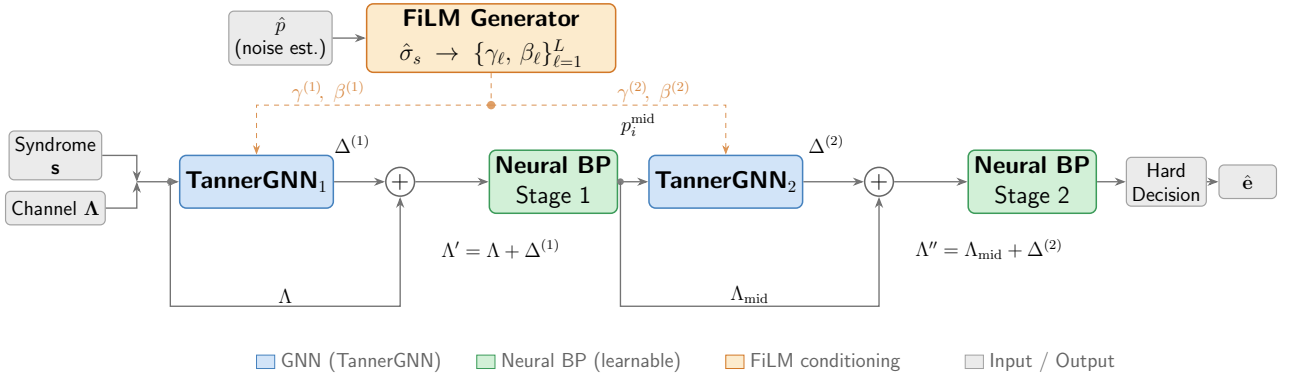


Figure 2: Interleaved GNN-BP pipeline. The GNN applies two successive LLR corrections: one before BP initialisation and one after T_1 iterations, using the mid-decoding posterior as an additional input. Both stages are trained end-to-end through the BP unroll.

trained on data regenerated each epoch with $p \sim \text{Uniform}[0.01, 0.10]$, and add a syndrome-consistency term, the binary cross-entropy between the measured syndrome and $\frac{1}{2}(1 - \prod_{j \in \mathcal{N}(c)} (1 - 2q_j))$ for each check c , with weight 0.1. Table 2 summarises the configuration.

4 Experimental Setup

4.1 Evaluation regimes

Two distinct evaluation regimes are used throughout this paper, and the distinction is central to interpreting the results:

Regime A (training-aligned baseline): The GNN is evaluated against flooding BP, the base decoder used during training. This comparison isolates the gain relative to the decoder represented in the training computation graph.

Regime B (stronger classical baseline): The same learned correction is evaluated with the serial-schedule min-sum BP-OSD reference configuration. This tests whether the correction also improves the stronger classical decoder considered here; Sec. 5.4 shows that it does not in the tested regime.

Unless otherwise stated, all experiments use Z -biased noise with $\eta = 20$.

4.2 Statistical protocol

Logical failure. For each shot let $\hat{e}_z$ and $\hat{e}_x$ be the decoder's hard decisions and $r_z = e_z \oplus \hat{e}_z$, $r_x = e_x \oplus \hat{e}_x$ the residual errors. A shot is a logical failure if (i) the estimate does not reproduce the measured syndrome, $\mathbf{H}_x \hat{e}_z \neq s_x$ or $\mathbf{H}_z \hat{e}_x \neq s_z$, or (ii) the syndrome is reproduced but the residual acts nontrivially on the code space, $\mathbf{L}_x r_z \neq \mathbf{0}$ or $\mathbf{L}_z r_x \neq \mathbf{0}$ over $\mathbb{F}_2$, where the rows

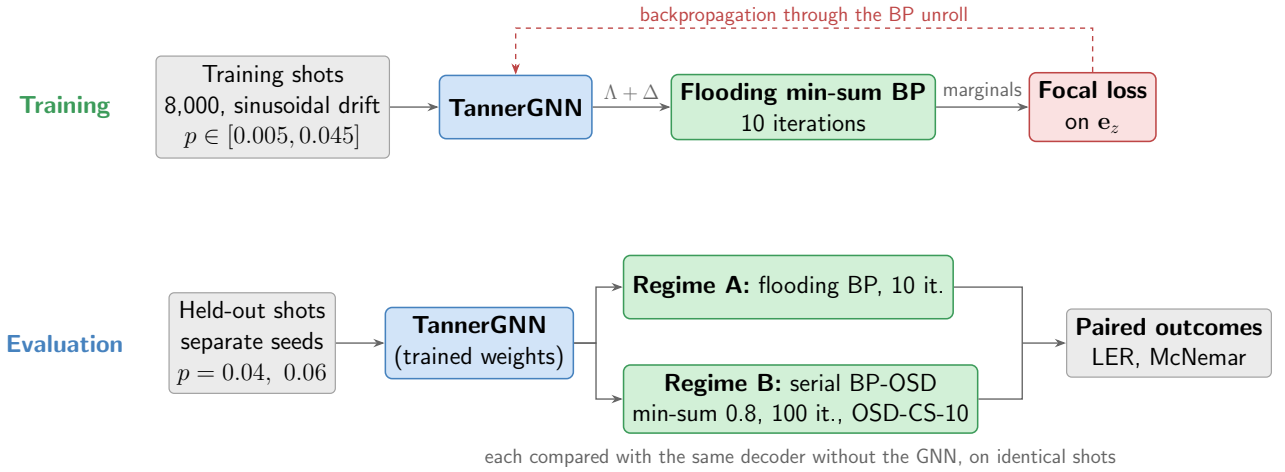


Figure 3: ~~Experimental Training and evaluation pipeline of the GNN experiments. Syndromes are sampled under static or drift noise models~~The GNN is trained through 10-iteration flooding BP and evaluated on separately seeded held-out shots in Regime A (flooding BP) and Regime B (serial BP-OSD); each decoder receives identical syndrome batches to enable McNemar paired testing.~~Oracle experiments substitute true per-qubit rates for the uniform mean prior.~~

of $\mathbf{L}_x$ ($\mathbf{L}_z$) span the $k = 12$ independent logical operators that detect Z (X) errors. Condition (i) is needed because a syndrome-inconsistent residual is not a logical operator, so its parity against fixed logical representatives is not meaningful; non-converged BP outputs therefore always count as failures. The X and Z components are combined by logical OR, so the reported LER is the probability that at least one of the 12 logical qubits suffers a logical error of any Pauli type in a shot. At circuit level the same rule is applied to the detector error model: a shot fails if the decoded fault set does not reproduce the detection events, or if it does and predicts the wrong observable flips.

For absolute LER comparisons we treat operating points with fewer than 100 observed logical-error events as preliminary; this is a reporting heuristic rather than a formal power calculation. Decoder pairs are evaluated on *identical* syndrome samples, enabling paired McNemar tests [13]. We report Wilson score 95% confidence intervals for LER estimates.

~~Random seeds are fixed at $s = 42$ for syndrome generation and $s + 999$ for validation splits. Unless otherwise stated, the reported Training, validation and test data are drawn with distinct random seeds, so no evaluation shot shares a random stream with a training shot, and learned-decoder result is based on a single training seed; its confidence interval and McNemar test~~

~~therefore quantify test-set sampling uncertainty conditional on that trained checkpoint, not training-to-training variability~~results are reported over five training seeds (initialisation and minibatch order). All training and evaluation used CPU only; no GPU was available for this work. ~~Representative per-epoch training times are ~ 17 s (154k-parameter GNN-BP) and ~ 37 s (183k-parameter interleaved model).~~One training epoch of the 39k-parameter model of Sec. 5.3 takes about 16 s.

5 Results

5.1 Decoder configuration sensitivity

Before presenting the GNN results, we quantify the sensitivity of the classical baseline to decoder configuration. This sensitivity sets the scale against which the learned-decoder results must be interpreted.

Table 3 illustrates how strongly decoder configuration changes the finite-length result. ~~In the 5,000-shot pilot, the~~The original parallel product-sum BP-OSD configuration reduces LER from 0.0506 to 0.0346 0.0569 to 0.0398 relative to the corresponding plain BP, but remains far worse than the serial min-sum configurations. The reference serial min-sum BP-OSD produces only one logical error in this sample, so the value 0.0002 should be interpreted as a preliminary estimate rather than a precisely

Table 2: Architecture and training hyperparameters configuration of the TANNERGNN model of Secs. 5.3 and 5.4.

GNN architecture	
Layers L / width d_h	3 / 32
Edge types	2 ($\mathbf{H}_x, \mathbf{H}_z$)
Node update	GRU, residual, layer norm
Node feature dim.	4
FiLM / attention	not used
Parameters	39,329
Training	
Data	8,000 shots, sinusoidal drift, $p \in [0.005, 0.045]$
Loss	focal, $(\alpha, \gamma) = (0.25, 2.0)$, on e_z ; no syndrome term
Optimizer	AdamW, lr 2×10^{-3} , weight decay 10^{-4}
Batch size / epochs	64 / 20, gradient clipping 1.0
Checkpoint	fewest failures on 4,000 validation shots
Training seeds	5
BP in training loop	flooding min-sum, scaling 0.8, 10 iter.
Reference BP-OSD baseline	
bp_method	ms (min-sum)
ms_scaling_factor	0.8
schedule	serial (layered)
max_iter	100
OSD (where used)	osd_cs_order 10

resolved LER records 34 logical errors in 10^5 shots, so its value is preliminary by the criterion of Sec. 4.2.

A separate $N = 100,000$ -shot experiment is shown in (Fig. 4. In that experiment, changing to) changes one factor at a time. The flooding comparator is min-sum BP with scaling factor 0.8, at most 100 iterations, stopping at the first iteration whose hard decision satisfies the syndrome, with a parallel (flooding) schedule; the second decoder is identical except for the serial schedule and then adding; the third adds OSD-CS-10 yields an overall approximately 40 $\times$ reduction relative to the flooding comparator. [State the exact BP method, scaling factor, iteration count, and stopping rule for the 100k flooding comparator.]. The LER falls from 1.28×10^{-2} to 1.45×10^{-3} (8.8 $\times$, paired $n_{10}/n_{01} = 1138/3$) and then to 2.7×10^{-4} (5.4 $\times$, 118/0), about 47 $\times$ overall.

These data do not identify a universally optimal configuration. They do show that schedule, update rule, iteration budget, and post-processing can dominate the measured finite-length LER. Learned-decoder comparisons must

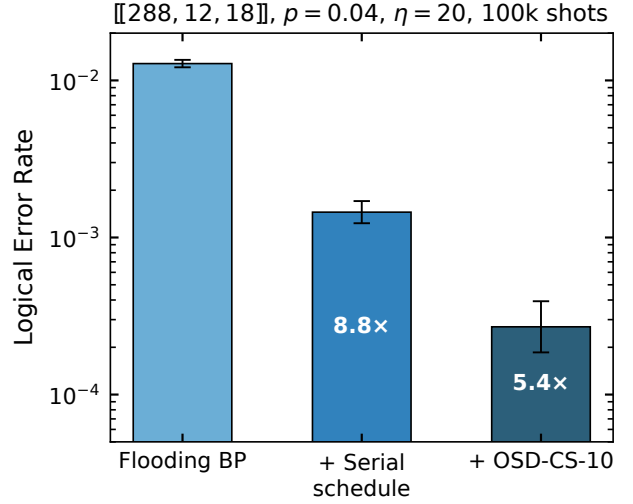


Figure 4: Baseline decomposition on $[[288, 12, 18]]$, $p = 0.04$, $\eta = 20$, 100k identical shots. The figure is intended to isolate the effect of serial scheduling Min-sum BP (scaling 0.8, at most 100 iterations) with flooding and subsequent OSD-CS-10 then serial schedule, then serial BP-OSD-CS-10. Wilson 95% CI error bars; labels give the exact flooding-BP comparator configuration must be reported in the text before the multiplicative factors are interpreted causally factor gained at each step.

therefore specify and control these parameters.

5.2 Uniform-prior scaling invariance of min-sum BP

Proposition 1 (Uniform positive-scaling invariance). *Consider separate-CSS min-sum BP with no message clipping or offset. If all channel LLRs in one CSS component are multiplied by the same factor $a > 0$, then every BP message and posterior LLR is multiplied by a at every iteration. Consequently, all hard decisions are unchanged.*

Proof. At initialization, each variable-to-check message is proportional to the corresponding channel LLR and therefore scales by a . Assume all variable-to-check messages at iteration t have scaled by a . The syndrome factor $(-1)^{s_j}$ in Eq. (3) is unchanged, the sign of every message is unchanged because $a > 0$, and $\min_i a|x_i| = a \min_i |x_i|$. Hence every check-to-variable message also scales by a . Equation (2) then shows that the next variable-to-check messages scale by a . Induction establishes the claim for all iterations and for the final posterior LLRs. $\square$

For a uniform channel prior, replacing a common positive LLR Λ by another positive value

Table 3: Effect of BP-OSD configuration on $[[288, 12, 18]]$, $p = 0.04$, $\eta = 20$, 5100,000 identical shots, priors p_z and p_x of Eq. (4). 95% Wilson intervals in brackets.

Config	Schedule	Method	Iter	LER	Errors
Original BP	parallel	prod-sum	30	0.0506-0.0569 [0.045,0.0570.0555,0.0583]	253-5,689
Reference BP	serial	ms(0.8)	100	0.0014-0.0017 [0.001,0.0030.0015,0.0020]	7-174
Original BP-OSD	parallel	prod-sum	30	0.0346-0.0398 [0.030,0.0400.0386,0.0411]	173-3,983
Ref. BP-OSD	serial	ms(0.8)	100	0.0002-0.00034 [0.000,0.0010.00024,0.00048]	1-34

Λ' is equivalent to the scaling $a = \Lambda'/\Lambda$. The proposition explains why changing only the assumed uniform physical error rate cannot change hard decisions of ideal min-sum BP. Clipping, off-sets, damping rules that depend on magnitude, non-uniform priors, or sum-product updates can break the exact scaling argument.

We observed the expected invariance in the implemented decoder: 1,740/1,740 bit-identical outputs when checked the proposition on 2,000 shots each. With unclipped min-sum BP (serial or flooding, 100 iterations), changing the assumed prior uniform rate from $p = 0.015$ to $p = 0.0286$ on $[[288, 12, 18]]$, and 1,000/1,000 bit-identical outputs for or from $p = 0.04$ versus to $p = 0.02$ in the tested on $[[72, 12, 6]]$, leaves every hard decision unchanged (2,000/2,000). Two cases fall outside the proposition, and the invariance fails there: our PyTorch min-sum setting. The same implementation clamps messages to ± 20 and changes 18 of 2,000-shot test was also identical for the tested 000 decisions on $[[288, 12, 18]]$, and sum-product configuration; because Proposition 1 does not apply to the tanh nonlinearity, we report that observation as empirical rather than structural. BP changes 25–43 of 2,000.

Remark 2 (What the proposition does and does not imply). The result applies to the BP pathway when adaptivity enters only by changing a uniform channel-prior magnitude. It does not establish that a global conditioning scalar is information-theoretically useless to a GNN: a learned model could combine a global noise estimate with local syndrome structure and change its nonlinear decision rule. The experiments below test a narrower hypothesis: spatially resolved per-qubit calibration can change the relative reliabilities seen by the decoder, whereas a single graph-wide scalar cannot provide that spatial information.

Table 4: GNN-BP vs. flooding BP on $[[72, 12, 6]]$, $\eta = 20$, 820,000 held-out test syndromes. GNN trained for 20 epochs on separate 8 ($p = 0.04$) and 10,000 training ($p = 0.06$) held-out test syndromes at $p \in [0.02, 0.03]$, five training seeds (mean $\pm$ s.d.). Best checkpoint at epoch 5 by validation LER d.). n_{10} (n_{01}): shots only BP (only GNN-BP) fails. BP is flooding min-sum.

	$p = 0.04$	$p = 0.06$
BP, 10 iter.	0.1077	0.2867
GNN-BP, 10 iter.	0.0942 $\pm$ 0.0006	0.2564 $\pm$ 0.0019
relative reduction	12.5% $\pm$ 0.6%	10.6% $\pm$ 0.7%
n_{10} / n_{01}	313–370 / 64–95	343–394 / 66–80
BP, 100 iter.	0.0863	0.2403
Serial BP-OSD	0.0707	0.2138

5.3 GNN improvement over flooding BP

We evaluated the GNN against flooding BP—the decoder it was trained against—on 8 separately seeded held-out syndromes: 4,000 held-out syndromes validation shots at $p = 0.04$ select the checkpoint over 20 epochs, and test sets of 20,000 shots at $p = 0.04$ (inside the training range) and 10,000 shots at $p = 0.06$ (outside it) are each scored once on that checkpoint. Training is repeated with five seeds.

Table 4 gives the primary learned-decoder result. At the epoch-5 checkpoint Over five training seeds, GNN-BP reduces LER by 15.4% relative to flooding BP on 8,000 held-out syndromes by 12.5% $\pm$ 0.6% at $p = 0.04$ ($n_{10} = 170$, $n_{01} = 48$, McNemar $\chi^2 = 67.16$, $p < 10^{-15}$). The test point lies and by 10.6% $\pm$ 0.7% at $p = 0.06$, outside the training range $p \in [0.02, 0.03]$, so the result demonstrates transfer to a higher physical error rate for this checkpoint. Because only one training seed is reported, we do not interpret the paired test as evidence for training-to-training reproducibility (McNemar $p < 10^{-36}$ for every seed). The gain comes mostly from convergence: shots on which BP does not reproduce the syndrome fall from 1,740 to 838–975 of 20,000.

Table 5: GNN-BP vs. serial BP-OSD on $[[72, 12, 6]]$, $\eta = 20$, 820,000 identical shots per p , five training seeds. Decoders evaluated on identical syndromes. Last column: seeds for which GNN+OSD is significantly worse (McNemar $p < 0.05$).

p	Serial BP-OSD	GNN+OSD	Seeds worse
0.020	0.0093	0.0095–0.0102	1/5
0.030	0.0297	0.0306–0.0317	5/5
0.040	0.0676	0.0688–0.0709	4/5
0.050	0.1354	0.1384–0.1418	5/5

Flooding BP with 100 iterations and serial BP-OSD both remain better than GNN-BP. A gradient audit found non-zero signal through the network (input-to-readout gradient ratio 0.22) and non-trivial LLR corrections (mean $|\Delta_i| = 0.854$).

A fixed-dataset capacity test provides an additional diagnostic: a GNN trained on 1,000 fixed syndromes for 50 epochs achieved a 14% LER reduction-reduced the failure count on those same syndromes (McNemar $p = 0.004$, $n_{10} = 17$ from 105 to 93 (11%; McNemar $p = 0.0095$, $n_{10} = 15$, $n_{01} = 3$). This experiment tests the ability of the architecture and optimizer to fit the supplied examples; it is not a generalization result. The interleaved model-reached-its-best-validation-checkpoint-at-epoch-1-and-did-not-train-successfully, so no performance conclusion about that architecture is drawn from the current run (see Limitations) architecture of Sec. 3.3 is not evaluated in this work.

Interleaved model training curves (183k parameters). The best validation checkpoint occurs at epoch 1 and the validation loss increases thereafter. The current run does not provide a trained interleaved decoder for comparison.

5.4 Training/evaluation baseline mismatch

Although the GNN improves flooding BP, the same correction does not improve the serial minimum BP-OSD reference configuration. Note that Tables 4 and 5 report different evaluation regimes (Regime A and Regime B of Sec. 4.1) on different syndrome sets, and their LER values are not directly comparable:

Table 5 and Fig. 5 show that the GNN-corrected decoder is statistically indistinguishable from never improves on serial BP-OSD at the two lower error rates and;

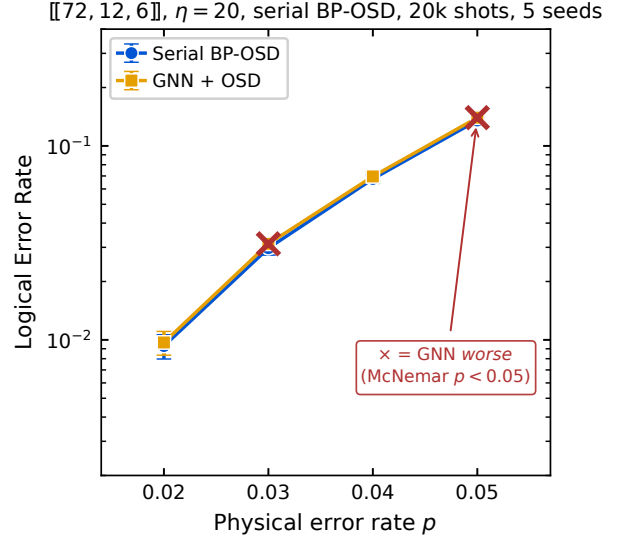


Figure 5: GNN+OSD vs. serial BP-OSD LER on $[[72, 12, 6]]$, $\eta = 20$, 820,000 shots per point; GNN+OSD is the mean over five training seeds. Red $\times$ marks operating points where GNN every seed is statistically worse (McNemar $p < 0.05$). Wilson 95% CI error bars.

at $p = 0.02$ one seed of five is significantly worse, and from $p = 0.03$ on 14 of 15 seed-error-rate pairs are significantly worse at the two higher rates. The same checkpoint improves flooding BP, so these data show that the learned correction does not transfer unchanged to the stronger decoder.

The experiment establishes a transfer failure, but it does not by itself identify the residual-failure mechanism. The GNN was optimized through a 10-iteration flooding-BP computation graph and is then inserted ahead of a decoder with a different schedule, iteration budget, and OSD post-processing. The experiment therefore does not establish that LLR corrections learned through the flooding-BP computation graph remain useful after the schedule, iteration budget, and post-processing are changed. At the same time, claims

A paired failure-set analysis (Table 6) shows that the two decoders fail on “structurally different” trapping sets require an explicit paired failure-overlap or trapping-set analysis, which is not included here. If available, add the Jaccard/overlap statistics of the two failure sets. do not fail on disjoint sets: every shot that 100-iteration BP fails is also failed by 10-iteration BP, and 93% of BP-OSD’s failures are BP failures. What separates them is mainly

Table 6: Paired failure sets on $[[72, 12, 6]]$, $p = 0.04$, $\eta = 20$, 20,000 identical held-out shots. BP: 10-iteration flooding BP. GNN rows give the range over five training seeds. “Non-conv.”: the estimate does not reproduce the syndrome.

Decoder	Failures	Non-conv.
BP, 10 iter.	2,155	1,740
BP, 100 iter.	1,727	700
Serial BP-OSD	1,414	0
GNN + BP, 10 iter.	1,874–1,906	838–975
GNN + BP-OSD	1,439–1,451	0
<i>Overlaps</i>		
BP-OSD failures also failed by BP	1,319 / 1,414 (93%)	
BP-100 failures also failed by BP	1,727 / 1,727 (100%)	
Shots GNN fixes for BP	313–370	
of which BP did not converge	91–93%	
of which BP-OSD also fails	31–36%	
BP-OSD failures also failed by GNN+BP	87–88%	
GNN+BP-OSD vs. BP-OSD: newly failed / rescued	237–282 / 206–254	

convergence: 81% of BP failures are shots on which BP does not reach a syndrome-consistent estimate, whereas BP-OSD always returns one. Of the 313–370 shots the GNN fixes for BP, 91–93% are non-converged BP shots and only 31–36% are shots that BP-OSD fails, so the GNN’s gains lie mostly where BP-OSD already succeeds. When the same corrections are passed to BP-OSD they reshuffle its failures (237–282 newly failed vs. 206–254 rescued shots), almost entirely among shots on which BP itself fails. The same pattern holds at $p = 0.06$, where 96% of BP-OSD failures are BP failures. Training directly against the target decoder, or against a differentiable surrogate of its post-processing stage, remains the appropriate test of whether learned augmentation can improve BP-OSD.

5.5 Per-qubit calibration gap

Under spatially non-uniform per-qubit noise ($p_i \sim \text{LogNormal}(\log p_{\text{mean}}, \sigma)$, fresh per shot), a each data qubit has rate $p_i = p e^{\sigma z_i - \sigma^2/2}$ with $z_i \sim \mathcal{N}(0, 1)$ drawn fresh every shot (capped at 0.5), so the mean rate is p for every σ . A decoder knowing the true per-qubit rates (serial min-sum BP-OSD-CS-10 with the true p_i as priors) substantially outperforms a mean-prior decoder decoder using the uniform mean prior p .

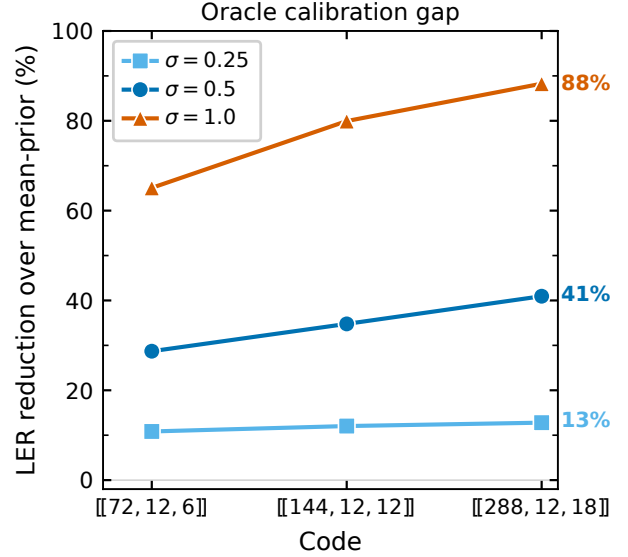


Figure 6: Oracle calibration gap (percentage LER reduction from per-qubit vs. mean-prior BP-OSD) at $\sigma = 1.0$, 50k identical shots ($[[72, 12, 6]]$, $[[144, 12, 12]]$) and 30k shots ($[[288, 12, 18]]$) per point. Operating points: $p = 0.04, 0.06, 0.07$ respectively; cross-code comparisons are partially confounded by the differing operating points.

Table 7 and Fig. 6 show a clear increase with noise contrast σ for each tested code. The reported percentage reduction is also larger for the larger-code rows, reaching 87%–88% on $[[288, 12, 18]]$ at $\sigma = 1.0$; however, the physical error rate is changed across codes ($p = 0.04, 0.06$, and 0.07). Consequently, these data do not isolate code distance as the cause of the cross-code trend.

This experiment isolates the value of synthetic per-qubit rate side information; it does not require machine learning. We use the true-rate decoder as an *oracle reference*, not as a formal upper bound on all possible decoders. The comparison quantifies how much the tested BP-OSD implementation can benefit when the realized per-qubit rates are revealed. The gap is *not* recoverable from a single syndrome shot under i.i.d. per-qubit noise: with a fresh field redrawn every shot, the uniform mean prior is Bayes-optimal within the class of estimators that observe only the syndrome, so no syndrome-conditioned GNN can do better than the mean prior in this regime. A GNN that attempts to estimate per-qubit rates from a single syndrome is significantly worse than the mean prior at $\sigma = 1.0$ (GNN LER = 0.069 vs. mean LER = 0.064, McNemar $p = 2.7 \times 10^{-4}$) not because

Table 7: Oracle gap: percentage LER reduction from per-qubit over mean-prior BP-OSD, 50,000 identical shots per cell, at $p = 0.04, 0.06, 0.07$ for the three codes. All McNemar $p \ll 10^{-3}$ $p < 10^{-20}$. Parenthetical counts are mean-prior error events.

Code (d)	$\sigma = 0.25$	$\sigma = 0.5$	$\sigma = 1.0$
$\llbracket 72, 12, 6 \rrbracket$ (6)	+11% (345 73508)	+30% (3420 +29% (3432)	+66% (2988 +65% (3438)
$\llbracket 144, 12, 12 \rrbracket$ (12)	+16% (276 +12% (3165)	+38% (268 +35% (3203)	+74% (2604 +80% (2936)
$\llbracket 288, 12, 18 \rrbracket$ (18)	+18% (922 +13% (3338)	+48% (524 +41% (3307)	+87% (1750 +88% (2786)

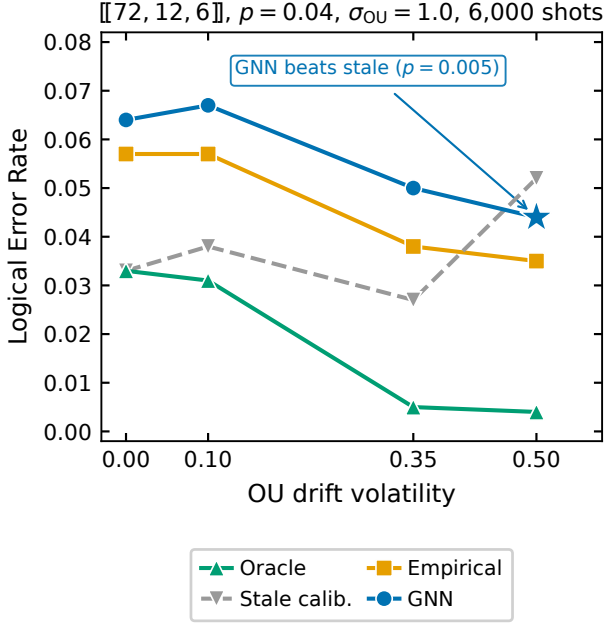


Figure 7: LER vs. OU drift volatility for four decoder configurations on $\llbracket 72, 12, 6 \rrbracket$, $p = 0.04$, $\sigma_{OU} = 1.0$, 6,000 shots. Star marks the single operating point (vol= 0.5) where GNN beats stale calibration ($p = 0.005$); GNN is beaten by the empirical frequency estimator at every volatility level.

~~the architecture is broken, but because the task is information-theoretically infeasible.~~

5.6 Drift adaptation

Under persistent OU drift ($\sigma = 1.0$, $W = 32$ -round window, $\llbracket 72, 12, 6 \rrbracket$, $p = 0.04$, 6,000 shots), the GNN beats stale calibration only at the highest tested drift volatility.

Table 8 and Fig. 7 show that the GNN beats stale calibration at vol= 0.5 ($p = 0.005$, 158 vs. 111 discordant pairs), but is beaten by the non-learned empirical frequency estimator at every volatility level (GNN vs. empir: $p = 2.7 \times 10^{-8}$ at vol= 0.5, GNN worse). Under independent per-qubit drift, direct per-qubit syndrome frequency counting captures most of the available

signal, leaving little room for a learned estimator. Whether spatial correlation structure between qubits would shift this result is an open question.

5.7 Circuit-level evaluation

~~[Specify the complete circuit-level noise model: syndrome ex~~

Circuit and noise model. We simulate a Z-basis memory experiment on $\llbracket 72, 12, 6 \rrbracket$ with 6 rounds of syndrome extraction (72 data qubits, 36 X-check and 36 Z-check ancillas). Data qubits are prepared in $|0\rangle$; each round measures all X-checks and then all Z-checks with dedicated ancillas, using CNOT layers from a greedy conflict-free schedule; the data qubits are finally measured in the Z basis. Detectors are (i) the 36 Z-checks of round 1, which are deterministic for the prepared state, (ii) the XOR of consecutive measurements of every check in rounds 2–6, and (iii) the XOR of the last Z-check outcome with the parity of the measured data qubits in its support, giving 432 detectors and 12 logical observables. Noise is a biased single-qubit Pauli channel with $p_X = p/(\eta + 1)$, $p_Z = p\eta/(\eta + 1)$, $p_Y = 0$, $\eta = 20$, applied to every qubit after each one- and two-qubit gate and to every idle qubit at each time step; measurements and resets flip with probability p . Under spatially non-uniform noise each qubit q (data and ancilla) has rate $p_q = p e^{\sigma z_q - \sigma^2/2}$, $z_q \sim \mathcal{N}(0, 1)$, $\sigma = 1$, on all of its fault locations, so the mean rate is p as in Sec. 5.5.

Decoders. We decode the detector error model (4,068 fault mechanisms) with serial min-sum BP-OSD (scaling 0.8, 100 iterations, OSD-CS order 10), as in Table 2. The mean-prior decoder uses the uniform rate p ; the oracle uses the true p_q . We draw 20 rate profiles at $p < 0.003$ and 40

Table 8: LER under per-qubit OU drift by volatility (vol). “Empir”: non-learned per-qubit frequency estimator.

vol	Oracle	Stale	Empir	GNN	Verdict
0.00	0.033	0.033	0.057	0.064	GNN worse, $p = 5 \times 10^{-36}$
0.10	0.031	0.038	0.057	0.067	GNN worse, $p = 8 \times 10^{-30}$
0.35	0.005	0.027	0.038	0.050	GNN worse, $p = 5 \times 10^{-19}$
0.50	0.004	0.052	0.035	0.044	GNN beats stale, $p = 0.005$

Circuit-level oracle gap on $[[72, 12, 6]]$, 6 rounds,
 $\eta = 20$, 5,000 shots per point (preliminary;
 $N = 5,000$ shots, underpowered).

Table 9: Circuit-level oracle gap on $[[72, 12, 6]]$ (6 rounds, $\eta = 20$, $\sigma = 1$). n_{10} (n_{01}): shots only the mean-prior decoder (only the oracle) fails. Zero failures in 10^4 shots means LER $< 3.8 \times 10^{-4}$ (95% upper bound).

p	Shots	Mean prior	Oracle	n_{10}/n_{01}	p_{McN}
0.001	10,000	0	0	0/0	—
0.003	10,000	0	0	0/0	—
0.01	20,000	1.35%	0.60%	177/28	5×10^{-25}

at $p = 0.01$, with 500 shots per profile. Failures follow Sec. 4.2.

Table 9 and Fig. 8 show no statistically detectable oracle gap at circuit level. With only 3 and 8 discordant pairs at the two tested operating points, however, McNemar’s test cannot distinguish a true zero gap from a gap of up to approximately 1.5% LER consistent with the code-capacity oracle gap at low contrast. The present sample cannot distinguish between negligible prior sensitivity and a nonzero gap below the resolution of the experiment. Higher shot counts (targeting ≥ 100 discordant pairs) are required to distinguish these cases. The two prior assignments are nevertheless numerically distinct: 86% of fault mechanisms change rank by more than 50 positions and the per-mechanism probability ratios span $0.25 \times 5.1 \times$. The DEM contains 3,821 fault mechanisms. Knowing the per-qubit rates also helps at circuit level (Table 9, Fig. 8). At $p = 0.01$ the oracle reduces the logical error rate from 1.35% to 0.60% (55%; McNemar $p = 4.8 \times 10^{-25}$), and it is better on 38 of 40 independent rate profiles (sign test $p = 1.5 \times 10^{-9}$), so the result is not driven by a few extreme profiles. At $p = 0.001$ and 360 detectors, suggesting that many fault mechanisms map to the same detector information. This high degeneracy is a plausible explanation for the weak prior sensitivity, but the

Circuit-level oracle gap on $[[72, 12, 6]]$, 6 rounds,
 $\eta = 20$, 5,000 shots. No statistically detectable difference between mean-prior and per-qubit-oracle BP-OSD at either operating point (McNemar $p \geq 0.25$; 3 and 8 discordant pairs—underpowered, see text).

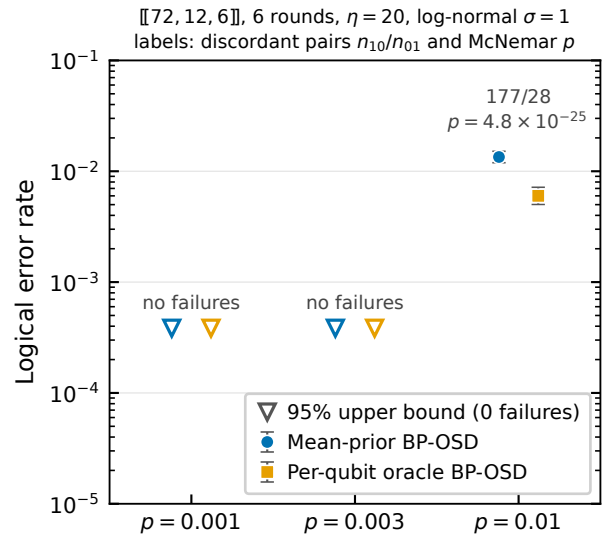


Figure 8: Circuit-level oracle gap on $[[72, 12, 6]]$, 6 rounds, $\eta = 20$, $\sigma = 1$. Points: LER with 95% Wilson intervals; open triangles: 95% upper bound when no failures were observed. Labels: discordant pairs and McNemar p .

present experiment does not isolate it causally. An ablation that changes DEM degeneracy while holding the prior mismatch fixed would be required for that conclusion.

Thus, the $p = 0.003$ neither decoder fails in 10^4 shots, so these operating points cannot resolve a gap. The circuit-level experiment is currently a null result with insufficient statistical resolution; gap is somewhat smaller than the code-capacity gap on the same code (65% at $\sigma = 1$, Table 7), at a different operating point, but it does not establish that calibration side information is irrelevant at circuit level.

6 Discussion

6.1 The GNN’s role and its limits

TANNERGNN ~~achieves a 15.4% LER reduction over~~ reduces the LER of the flooding BP decoder used in its training loop. ~~The paired held-out test supports a real difference for this trained checkpoint, while by $12.5\% \pm 0.6\%$ over five training seeds, and~~ gradient diagnostics show that the network produces non-trivial corrections. ~~Robustness to training initialization has not yet been measured, so we do not treat the single-seed result as a reproducibility claim~~ The gain comes mainly from helping BP converge, and 100-iteration BP and serial BP-OSD remain stronger than the corrected decoder.

The Regime-B result is also informative: the learned correction does not transfer automatically to serial BP-OSD. This is consistent with the training/evaluation mismatch, ~~but the residual failure sets have not yet been characterized well enough to claim a trapping-set mechanism: the failure sets of BP and BP-OSD are nested, and the GNN’s repairs fall mostly on shots BP-OSD already decodes (Table 6).~~ Training against BP-OSD directly is complicated by the non-differentiable OSD stage; differentiable surrogates, distillation, or training on the residual failures of the target decoder are possible alternatives.

6.2 Benchmark design implications

The baseline experiments identify two reporting requirements for learned QLDPC decoders. First, BP-OSD schedule, update rule, iteration count, and OSD configuration must be stated explicitly because they can change LER by more than two orders of magnitude on the same code. Table 2 records these settings for the present study.

Second, the training and evaluation decoders must be distinguished. An LLR correction learned through flooding BP need not remain beneficial when the schedule, iteration count, and OSD stage are changed. Comparisons across such decoder configurations should identify the mismatch rather than treating all BP-based baselines as equivalent.

6.3 Comparison with prior work

Astra [5] uses a learned Tanner-graph message-passing decoder and reports BB-code performance competitive with, and in several settings better than, BP-OSD under code-capacity depolarizing noise. This places our ~~15.4%~~ 12.5% flooding-BP gain in context: improvement over plain BP is not, by itself, the main novelty. Our contribution is instead the controlled comparison showing how that gain changes when the baseline and side information are strengthened.

Gong et al. [4] interleave differentiable BP stages with GNN corrections on the same sparse graph and show error-floor improvements over several post-processing approaches. Our interleaved variant follows the same broad design principle, but ~~the checkpoint reported here does not train successfully. We therefore its decoding performance is not evaluated here, so we~~ do not claim a competitive result for that architecture.

Recent circuit-level work on BB codes also reports neural decoders that can outperform BP-OSD on the $[[72, 12, 6]]$ code while retaining a runtime advantage, although scaling to $[[144, 12, 12]]$ remains more challenging [14]. This comparison reinforces the need to evaluate learned decoders against a fully specified competitive non-learned baseline, not only against flooding BP.

Stein et al. [15] condition a neural QEC decoder on spatially resolved hardware calibration data using FiLM and evaluate on IBM repetition-code experiments. Their conditioning signal is per-qubit and calibration-derived, unlike the single scalar $\hat{p}$ used in our initial model. This distinction is consistent with the oracle experiments of Sec. 5.5, which motivate spatial rather than purely global conditioning.

6.4 Towards spatially-resolved conditioning

The oracle-gap experiment (Sec. 5.5) shows that per-qubit rate side information can materially improve the tested BP-OSD decoder. Because operating point and code size co-vary, the current data do not establish how this value scales with distance. Proposition 1 also shows that a uniform rescaling of the min-sum BP prior cannot reproduce the effect of spatially varying reliabilities. These results motivate testing spatially resolved FiLM (SRCF), in which per-qubit parameters (γ_i, β_i) are conditioned on persistent hardware cal-

ibration data rather than on a single graph-wide scalar. Such a decoder should be compared with the non-learned frequency estimator of Sec. 5.6 and evaluated in a circuit-level setting where the calibration signal remains identifiable. Neither condition has been established in the present experiments.

6.5 Limitations

- (i) ~~The primary 15.4% GNN result uses one training seed. Statistical uncertainty over independent training runs has not been measured.~~
- (ii) All GNN variants were trained against 10-iteration flooding BP. No variant was trained against serial BP-OSD, which is the competitive classical baseline. Whether GNN augmentation of serial BP-OSD yields gains is untested.
- (iii) GNN checkpoints exist only for $[[72, 12, 6]]$; the larger codes were not trained due to CPU-only compute constraints. The null result in Regime B should not be extrapolated to the larger codes.
- (iv) The interleaved ~~model reached its best validation checkpoint at epoch 1, after which the validation loss increased. The current run is therefore insufficient to evaluate the interleaved architecture and~~ FiLM architectures are described but their decoding performance is not evaluated.
- (v) All training and evaluation used CPU only, which limited hyperparameter search and training duration.
- (vi) The single per-qubit correction Δ_i is applied to both $\Lambda_{z,i}$ and $\Lambda_{x,i}$, which is suboptimal under $\eta = 20$ biased noise; separate Z/X readout heads are the recommended fix.
- (vii) Code-capacity noise omits measurement errors, leakage, and correlated faults present in real circuits.

7 Conclusion

On $[[72, 12, 6]]$, a Tanner-graph GNN trained through 10-iteration flooding BP reduces the held-out LER ~~by 15.4% of that decoder by~~

$12.5\% \pm 0.6\%$ at $p = 0.04$ and $\eta = 20$ over five training seeds. The same learned correction does not improve the serial min-sum BP-OSD reference decoder and ~~increases LER at the two highest tested error rates~~ is significantly worse from $p = 0.03$ on; the shots it repairs are largely ones BP-OSD already decodes. The demonstrated gain is therefore specific to the training-aligned flooding-BP baseline; the present experiments do not establish an improvement over competitive BB decoding more generally.

The baseline experiments show that BP schedule, update rule, iteration budget, and OSD configuration can change finite-length LER by orders of magnitude. These parameters must be treated as part of the decoder specification when learned and non-learned methods are compared. We also proved that uniform positive rescaling of the channel LLRs leaves the hard decisions of ideal min-sum BP unchanged. A global prior magnitude alone therefore cannot provide spatial adaptation to heterogeneous qubit reliabilities, although a nonlinear learned model may still use a global noise estimate as context.

The synthetic oracle experiments identify per-qubit calibration as the more informative adaptive signal in the tested setting: revealing the true per-qubit rates substantially improves BP-OSD under heterogeneous noise, and under circuit-level noise it reduces BP-OSD's LER by 55% at $p = 0.01$. The current cross-code experiments do not isolate how this gain scales with distance, ~~and the circuit-level study does not yet have sufficient statistical resolution.~~ A direct next test is to condition a learned decoder on persistent spatial calibration data and compare it against a fully specified serial BP-OSD baseline and the non-learned frequency estimator, using multiple training seeds and a noise model that preserves temporal calibration structure.

Acknowledgments

The authors thank the STAR Lab program at the University of Arizona for facilitating this research and providing computational resources.

Data and code availability

Source code, trained checkpoints, ~~all datasets, and the audit scripts that produced the~~

~~findings of Secs. 5.1 and 5.2 datasets and the analysis scripts behind every table and figure except the drift experiment (Sec. 5.6)~~ are available at ~~. Syndrome data were generated using~~ https://github.com/kondshk/adaptive_gnn. ~~Code-capacity data are sampled directly with NumPy; circuit-level data are generated with Stim [16].~~

References

- [1] Sergey Bravyi, Andrew W. Cross, Jay M. Gambetta, Dmitri Maslov, Patrick Rall, and Theodore J. Yoder. “High-threshold and low-overhead fault-tolerant quantum memory”. *Nature* **627**, 778–782 (2024).
- [2] IBM Quantum. “IBM lays out clear path to fault-tolerant quantum computing”. IBM Quantum Computing Blog (2025).
- [3] Joschka Roffe, David R. White, Simon Burton, and Earl Campbell. “Decoding across the quantum LDPC code landscape”. *Physical Review Research* **2**, 043423 (2020).
- [4] Anqi Gong, Sebastian Cammerer, and Joseph M. Renes. “Graph neural networks for enhanced decoding of quantum LDPC codes”. In Proc. IEEE International Symposium on Information Theory (ISIT). (2024).
- [5] Arshpreet Singh Maan and Alexandru Paler. “Machine learning message-passing for the scalable decoding of QLDPC codes”. *npj Quantum Information* **11**, 78 (2025).
- [6] A. Robert Calderbank and Peter W. Shor. “Good quantum error-correcting codes exist”. *Physical Review A* **54**, 1098 (1996).
- [7] Andrew M. Steane. “Error correcting codes in quantum theory”. *Physical Review Letters* **77**, 793 (1996).
- [8] Marc P. C. Fossorier and Shu Lin. “Soft-decision decoding of linear block codes based on ordered statistics”. *IEEE Transactions on Information Theory* **41**, 1379–1396 (1995).
- [9] P. V. Klimov et al. “Fluctuations of energy-relaxation times in superconducting qubits”. *Physical Review Letters* **121**, 090502 (2018).
- [10] Malcolm Carroll, Sami Rosenblatt, Petar Jurcevic, Isaac Lauer, and Abhinav Kandala. “Dynamics of superconducting qubit relaxation times”. *npj Quantum Information* **8**, 132 (2022). [arXiv:2105.15201](https://arxiv.org/abs/2105.15201).
- [11] Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville. “FiLM: Visual reasoning with a general conditioning layer”. In Proc. AAAI. Volume 32. (2018). [arXiv:1709.07871](https://arxiv.org/abs/1709.07871).
- [12] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. “Focal loss for dense object detection”. In Proc. IEEE ICCV. Pages 2999–3007. (2017).
- [13] Quinn McNemar. “Note on the sampling error of the difference between correlated proportions or percentages”. *Psychometrika* **12**, 153–157 (1947).
- [14] John Blue, Harshil Avlani, Zhiyang He, Liu Ziyin, and Isaac L. Chuang. “Machine learning decoding of circuit-level noise for bivariate bicycle codes”. *Quantum* **10**, 2149 (2026). [arXiv:2504.13043](https://arxiv.org/abs/2504.13043).
- [15] Samuel Stein, Shuwen Kan, Chenxu Liu, Adrian Harkness, Sean Garner, Zefan Du, Yufei Ding, Ying Mao, and Ang Li. “Calibration-conditioned featurewise linear modulation decoders for low-latency decoding of quantum error correction evaluated on IBM repetition-code experiments”. *Physical Review Research* **8**, 033194 (2026). [arXiv:2601.16123](https://arxiv.org/abs/2601.16123).
- [16] Craig Gidney. “Stim: A fast stabilizer circuit simulator”. *Quantum* **5**, 497 (2021).